

AGREEGATE FUNCTIONS

Geethika (2520030485)

Question :

Perform SQL aggregate operations on bank transaction data using SUM(), AVG(), MAX(), MIN(), and COUNT() along with WHERE, GROUP BY, HAVING, and ORDER BY clauses to analyse transaction amounts, branches, deposits, withdrawals, and transaction counts.

Step 1 : Create Database and Table

The screenshot displays the SQL Server Enterprise Manager interface. The left pane shows the 'SCHEMAS' tree with a filter on 'Filter objects'. The right pane shows the 'SQL File 3' script editor with the following SQL code:

```
1 CREATE DATABASE aggregate_bank_db;
2
3 USE aggregate_bank_db;
4
5 CREATE TABLE bank_transactions (
6     txn_id INT PRIMARY KEY,
7     customer_name VARCHAR(50),
8     branch_name VARCHAR(50),
9     transaction_type VARCHAR(20),
10    amount DECIMAL(10,2),
11    transaction_date DATE
12 );
```

The bottom pane shows the 'Output' window with the following results:

#	Time	Action	Message
1	18:14:33	CREATE DATABASE aggregate_bank_db	1 row(s) affected
2	18:14:33	USE aggregate_bank_db	0 row(s) affected
3	18:14:33	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VA...	0 row(s) affected

30485 x

File Edit View Query Database Server Tools Scripting Help

Navigator SQL File 3* Limit to 1000 rows

SCHEMAS

Filter objects

- bank_management_db
- bookflow_db
- joins_operations_db
 - Tables
 - Views
 - Stored Procedures
 - Functions
- std_marks
- sys

Administration Schemas

Information

No object selected

```

8  txn_id INT PRIMARY KEY,
9  customer_name VARCHAR(50),
10 branch_name VARCHAR(50),
11 transaction_type VARCHAR(20),
12 amount DECIMAL(10,2),
13 transaction_date DATE
14 );
15
16 • INSERT INTO bank_transactions VALUES
17 (101, 'Geethika', 'Hyderabad', 'Deposit', 5000, '2024-01-05'),
18 (102, 'Shailu', 'Hyderabad', 'Withdrawal', 2000, '2024-01-06'),
19 (103, 'Jhansi', 'Vijayawada', 'Deposit', 12000, '2024-01-08'),
20 (104, 'Siri', 'Vizag', 'Deposit', 8000, '2024-01-10'),
21 (105, 'Aishu', 'Hyderabad', 'Withdrawal', 3500, '2024-01-11'),
22 (106, 'Bhavitha', 'Vizag', 'Deposit', 15000, '2024-01-12'),
23 (107, 'Naksh', 'Vijayawada', 'Withdrawal', 1000, '2024-01-13'),
24 (108, 'Mahija', 'Hyderabad', 'Deposit', 9000, '2024-01-14'),
25 (109, 'Prakshitha', 'Vizag', 'Withdrawal', 4000, '2024-01-15'),
26 (110, 'Pavani', 'Vijayawada', 'Deposit', 11000, '2024-01-16');

```

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Context Help Snippets

Output

Action Output

#	Time	Action	Message
6	18:17:13	CREATE DATABASE aggregate_bank_db	1 row(s) affected
7	18:17:13	USE aggregate_bank_db	0 row(s) affected
8	18:17:13	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name V...	0 row(s) affected
9	18:17:13	INSERT INTO bank_transactions VALUES (101, 'Geethika', 'Hyderabad', 'Deposit', ...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0

Object Info Session

Step 2 : AGREEEGATE FUNCTIONS

30485 x

File Edit View Query Database Server Tools Scripting Help

Navigator SQL File 3* Limit to 1000 rows

SCHEMAS

Filter objects

- bank_management_db
- bookflow_db
- joins_operations_db
 - Tables
 - Views
 - Stored Procedures
 - Functions
- std_marks
- sys

Administration Schemas

Information

No object selected

```

21 (105, 'Aishu', 'Hyderabad', 'Withdrawal', 3500, '2024-01-11'),
22 (106, 'Bhavitha', 'Vizag', 'Deposit', 15000, '2024-01-12'),
23 (107, 'Naksh', 'Vijayawada', 'Withdrawal', 1000, '2024-01-13'),
24 (108, 'Mahija', 'Hyderabad', 'Deposit', 9000, '2024-01-14'),
25 (109, 'Prakshitha', 'Vizag', 'Withdrawal', 4000, '2024-01-15'),
26 (110, 'Pavani', 'Vijayawada', 'Deposit', 11000, '2024-01-16');
27
28 -- Step 3: SUM()
29 • SELECT SUM(amount) AS Total_Amount
30 FROM bank_transactions;

```

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or to toggle a

Result Grid

Filter Rows:	Export:	Wrap Cell Content:
Total_Amount		
70500.00		

Result 1 x

Output

Action Output

#	Time	Action	Message
12	18:17:52	USE aggregate_bank_db	
13	18:17:52	CREATE TABLE bank_transactions (txn_id	
14	18:17:52	INSERT INTO bank_transactions VALUES	
15	18:17:52	SELECT SUM(amount) AS Total_Amount FF	10 Duplicates: 0 Warnings: 0

Object Info Session

File Edit View Query Database Server Tools Scripting Help

Navigator: SQL File 3* x

SCHEMAS

Filter objects

- bank_management_db
- bookflow_db
- joins_operations_db
 - Tables
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 - Functions
- std_marks
- sys

Administration Schemas

Information

No object selected

SQL File 3* x

```

25 (100, 'Prakshitha', 'Vizag', 'Withdrawal', 4000, '2024-01-15'),
26 Execute the selected portion of the script or everything, if there is no selection
27
28 -- Step 3: SUM()
29 • SELECT SUM(amount) AS Total_Amount
30 FROM bank_transactions;
31
32 -- Step 4: AVG()
33 • SELECT AVG(amount) AS Average_Transaction
34 FROM bank_transactions;

```

Result Grid

Column	Value
Average_Transaction	7050.000000

Result 2 Result 3 x

Read Only Context Help

Output

Action Output

#	Time	Action	Message
✓ 19	18:18:18	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name ...	0 row(s) affected
✓ 20	18:18:18	INSERT INTO bank_transactions VALUES (101, 'Geethika', 'Hyderabad', 'Deposit', ...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
✓ 21	18:18:18	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
✓ 22	18:18:18	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT ...	1 row(s) returned

Object Info Session

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File Edit View Query Database Server Tools Scripting Help

Navigator: SQL File 3* x

SCHEMAS

Filter objects

- bank_management_db
- bookflow_db
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 - Tables
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 - Stored Procedures
 - Functions
- std_marks
- sys

Administration Schemas

Information

No object selected

SQL File 3* x

```

32 -- Step 4: AVG()
33 • SELECT AVG(amount) AS Average_Transaction
34 FROM bank_transactions;
35
36 SELECT MAX(amount) AS Highest_Transaction
37 FROM bank_transactions;
38
39 -- Step 6: MIN()
40 • SELECT MIN(amount) AS Lowest_Transaction
41 FROM bank_transactions;

```

Result Grid

Column	Value
Lowest_Transaction	1000.00

Result 4 Result 5 Result 6 Result 7 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration
✓ 28	18:19:01	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.000 se
✓ 29	18:19:01	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT ...	1 row(s) returned	0.000 se
✓ 30	18:19:01	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions LIMIT 0...	1 row(s) returned	0.000 se
✓ 31	18:19:01	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, ...	1 row(s) returned	0.000 se

Object Info Session

Automatic context help is
Use the toolbar to manu
help for the current caret
or to toggle automatic

File Edit View Query Database Server Tools Scripting Help

SQL File 3*

Limit to 1000 rows

```

36 SELECT MAX(amount) AS Highest_Transaction
37 FROM bank_transactions;
38
39 -- Step 6: MIN()
40 • SELECT MIN(amount) AS Lowest_Transaction
41 FROM bank_transactions;
42
43 -- Step 7: COUNT()
44 • SELECT COUNT(*) AS Total_Transactions
45 FROM bank_transactions;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Contents: |

Total_Transactions
10

Administration Schemas

Information

No object selected

Object Info Session

Result 8 Result 9 Result 10 Result 11 Result 12 x Read Only

Output

Action Output

#	Time	Action	Message
✓ 38	18:19:24	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT ...	1 row(s) returned
✓ 39	18:19:24	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions LIMIT 0...	1 row(s) returned
✓ 40	18:19:24	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, ...	1 row(s) returned
✓ 41	18:19:24	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000	1 row(s) returned

Step 3 : WHERE + AGGREGATE

File Edit View Query Database Server Tools Scripting Help

Navigator SQL File 3* x

Limit to 1000 rows

41 FROM bank_transactions;
42
43 -- Step 7: COUNT()
44 • SELECT COUNT(*) AS Total_Transactions
45 FROM bank_transactions;
46
47 -- Step 8: WHERE + SUM()
48 • SELECT SUM(amount) AS Total_Deposit
49 FROM bank_transactions
50 WHERE transaction_type = 'Deposit';

Result Grid

Total_Deposit
60000.00

Administration Schemas

Information

No object selected

Result 13 Result 14 Result 15 Result 16 Result 17 Result 18 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration /
49	18:20:05	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions LIMIT 0...	1 row(s) returned	0.000 sec
50	18:20:05	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions	returned	0.000 sec
51	18:20:05	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions	returned	0.000 sec
52	18:20:05	SELECT SUM(amount) AS Total_Deposit FROM bank_transactions	returned	0.000 sec

Object Info Session

Automatic context help is displayed. Use the toolbar to manually toggle automatic help for the current caret position.

Step 4 : GROUP BY

The screenshot shows the SQL Developer interface with a SQL script editor and a results grid. The script contains two queries: Step 8, which calculates the total deposit, and Step 9, which calculates the total amount by branch. The results grid displays the output of Step 9, showing the total amount for three branches: Hyderabad (19500.00), Vijayawada (24000.00), and Vizag (27000.00). The bottom pane shows the output of the queries, including the execution time and the number of rows returned.

SQL Script:

```
-- Step 8: WHERE + SUM()
47
48 • SELECT SUM(amount) AS Total_Deposit
49 FROM bank_transactions
50 WHERE transaction_type = 'Deposit';
51
52 -- Step 9: GROUP BY + SUM()
53 • SELECT branch_name,
54 SUM(amount) AS Total_Amount
55 FROM bank_transactions
56 GROUP BY branch_name;
```

Results Grid:

branch_name	Total_Amount
Hyderabad	19500.00
Vijayawada	24000.00
Vizag	27000.00

Output:

#	Time	Action	Message
61	18:20:29	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, ...	1 row(s) returned
62	18:20:29	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
63	18:20:29	SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE trans...	1 row(s) returned
64	18:20:29	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions...	3 row(s) returned

Step 5 : GROUP BY + HAVING

30485 x

File Edit View Query Database Server Tools Scripting Help

SQL File 3*

Limit to 1000 rows

SQLAdditions

Jump to

Automatic context help is available. Use the toolbar to manage help for the current caret position or to toggle automatic help.

SCHEMAS

Filter objects

- bank_management_db
- bookflow_db
- joins_operations_db
 - Tables
 - Views
 - Stored Procedures
 - Functions
- std_marks
- sys

Administration Schemas

Information

No object selected

```

54 SUM(amount) AS Total_Amount
55 FROM bank_transactions
56 GROUP BY branch_name;
57
58 -- Step 10: GROUP BY + HAVING
59 SELECT branch_name,
60 SUM(amount) AS Total_Amount
61 FROM bank_transactions
62 GROUP BY branch_name
63 HAVING SUM(amount) > 20000;

```

Result Grid

branch_name	Total_Amount
Vijayawada	24000.00
Vizag	27000.00

Result 26 Result 27 Result 28 Result 29 Result 30 Result 31 Result 32 Result 33 x Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration
74	18:21:18	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.000
75	18:21:18	SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE trans...	1 row(s) returned	0.000
76	18:21:18	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions...	3 row(s) returned	0.000
77	18:21:18	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions...	2 row(s) returned	0.000

Step 6 : GROUP BY + ORDER BY

30485 x

File Edit View Query Database Server Tools Scripting Help

Navigator: SQL File 3* x

SCHEMAS

Filter objects

- bank_management_db
- bookflow_db
- joins_operations_db
 - Tables
 - Views
 - Stored Procedures
 - Functions
- std_marks
- sys

Administration Schemas

Information

No object selected

```

61 FROM bank_transactions
62 GROUP BY branch_name
63 HAVING SUM(amount) > 20000;
64
65 -- Step 11: GROUP BY + ORDER BY
66 • SELECT branch_name,
67 SUM(amount) AS Total_Amount
68 FROM bank_transactions
69 GROUP BY branch_name
70 ORDER BY Total_Amount DESC;

```

Result Grid

branch_name	Total_Amount
Vizag	27000.00
Vijayawada	24000.00
Hyderabad	19500.00

Export: Wrap Cell Content: [F1](#)

Result 34 Result 35 Result 36 Result 37 Result 38 Result 39 Result 40 Result 41 Result 42 Read Only Context

Output

Action Output

#	Time	Action	Message
88	18:21:53	SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE trans...	1 row(s) returned
89	18:21:53	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions...	3 row(s) returned
90	18:21:53	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions...	2 row(s) returned
91	18:21:53	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions...	3 row(s) returned

Object Info Session

Step 7: GROUP BY + HAVING + ORDER BY

MySQL Workbench

30485 x

File Edit View Query Database Server Tools Scripting Help

Navigator: SCHEMAS

Filter objects

- bank_management_db
- bookflow_db
- joins_operations_db
 - Tables
 - Views
 - Stored Procedures
 - Functions
- std_marks
- sys

SQL File 3* x

Limit to 1000 rows

```
69 GROUP BY branch_name
70 ORDER BY Total_Amount DESC;
71
72 -- Step 12: GROUP BY + HAVING + ORDER BY
73 • SELECT branch_name,
74 COUNT(*) AS Total_Transactions
75 FROM bank_transactions
76 GROUP BY branch_name
77 HAVING COUNT(*) >= 3
78 ORDER BY Total_Transactions DESC;
```

Result Grid

branch_name	Total_Transactions
Hyderabad	4
Vijayawada	3
Vizag	3

Administration Schemas

Information

No object selected

Result 44 Result 45 Result 46 Result 47 Result 48 Result 49 Result 50 Result 51 Result 52 Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration
103	18:22:52	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions...	3 row(s) returned	0.000 sec
104	18:22:52	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions...	2 row(s) returned	0.000 sec
105	18:22:52	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions...	3 row(s) returned	0.000 sec
106	18:22:52	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactio...	3 row(s) returned	0.000 sec

Object Info Session

MySQL Workbench

30485 x

File Edit View Query Database Server Tools Scripting Help

Navigator: SCHEMAS

Filter objects

- bank_management_db
- bookflow_db
- joins_operations_db
 - Tables
 - Views
 - Stored Procedures
 - Functions
- std_marks
- sys

SQL File 3* x

Limit to 1000 rows

```
74 COUNT(*) AS Total_Transactions
75 FROM bank_transactions
76 GROUP BY branch_name
77 HAVING COUNT(*) >= 3
78 ORDER BY Total_Transactions DESC;
79
80 -- Step 13: WHERE + COUNT()
81 • SELECT COUNT(*) AS Withdrawals
82 FROM bank_transactions
83 WHERE transaction_type = 'Withdrawal';
```

Result Grid

Withdrawals
4

Administration Schemas

Information

No object selected

Result 55 Result 56 Result 57 Result 58 Result 59 Result 60 Result 61 Result 62 Result 63 Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration
119	18:23:49	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions...	2 row(s) returned	0.000 sec
120	18:23:49	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions...	3 row(s) returned	0.000 sec
121	18:23:49	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactio...	3 row(s) returned	0.000 sec
122	18:23:49	SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transactio...	1 row(s) returned	0.000 sec

Object Info Session

Step 9 : WHERE + AVG()

30485 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bank_management_db
- bookflow_db
- joins_operations_db
 - Tables
 - Views
 - Stored Procedures
 - Functions
- std_marks
- sys

SQL File 3*

Limit to 1000 rows

```

79
80 -- Step 13: WHERE + COUNT()
81 • SELECT COUNT(*) AS Withdrawals
82 FROM bank_transactions
83 WHERE transaction_type = 'Withdrawal';
84
85 -- Step 14: WHERE + AVG()
86 • SELECT AVG(amount) AS Avg_Deposit
87 FROM bank_transactions
88 WHERE transaction_type = 'Deposit';

```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

Avg_Deposit
10000.000000

Administration Schemas

Information

No object selected

Result 67 Result 68 Result 69 Result 70 Result 71 Result 72 Result 73 Result 74 Result 75 Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
136	18:24:14	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions...	3 row(s) returned	0.000 sec / 0.00
137	18:24:14	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions...	3 row(s) returned	0.016 sec / 0.00
138	18:24:14	SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type = 'Withdrawal'...	1 row(s) returned	0.000 sec / 0.00
139	18:24:14	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit'...	1 row(s) returned	0.000 sec / 0.00

Object Info Sessions

```

75 -- Step 15: GROUP BY + MAX()
76 • SELECT branch_name,
77 MAX(amount) AS Highest_Amount
78 FROM bank_transactions
79 GROUP BY branch_name;

```

Result Grid

branch_name	Highest_Amount
Hyderabad	9000.00
Vijayawada	12000.00
Vizag	15000.00

Result 14 x

Output

Action Output

#	Time	Action	Message
15	21:34:19	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name...	2 row(s) returned
16	21:35:45	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name...	3 row(s) returned
17	21:36:46	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_name...	3 row(s) returned
18	21:37:48	SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type = 'Withdrawal' LIMIT ...	1 row(s) returned
19	21:38:41	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT ...	1 row(s) returned
20	22:08:34	SELECT branch_name, MAX(amount) AS Highest_Amount FROM bank_transactions GROUP BY branch_name...	3 row(s) returned

Step 11: GROUP BY + MIN()

```

80  -- Step 16: GROUP BY + MIN()
81  • SELECT branch_name,
82         MIN(amount) AS Lowest_Amount
83  FROM bank_transactions
84  GROUP BY branch_name;

```

branch_name	Lowest_Amount
Hyderabad	2000.00
Vijayawada	1000.00
Vizag	4000.00

Result 15 x

Output

Action Output

#	Time	Action	Message
✓ 17	21:36:46	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_...	3 row(s) returned
✓ 18	21:37:48	SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type = 'Withdrawal' LIMIT ...	1 row(s) returned
✓ 19	21:38:41	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT...	1 row(s) returned
✓ 20	22:08:34	SELECT branch_name, MAX(amount) AS Highest_Amount FROM bank_transactions GROUP BY branch...	3 row(s) returned
✗ 21	22:09:54	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions LIMIT 0, 1000	Error Code: 1140. In
✓ 22	22:10:54	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions GROUP BY branch_...	3 row(s) returned

Step 12: WHERE + ORDER BY

MySQL Workbench

30485 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bank_management_db
- bookflow_db
- joins_operations_db
 - Tables
 - Views
 - Stored Procedures
 - Functions
- std_marks
- sys

SQL File 3" x

```

127  Step 21: WHERE + GROUP BY + HAVING
128  SELECT branch_name,
129         SUM(amount) AS Total_Deposit
130  FROM bank_transactions
131  WHERE transaction_type = 'Deposit'
132  GROUP BY branch_name
133  HAVING SUM(amount) > 15000;
134
135  -- Step 22: GROUP BY + HAVING + ORDER BY
136  • SELECT branch_name,
137         AVG(amount) AS Avg_Amount

```

branch_name	Total_Transactions
Hyderabad	4

Result Grid

Form Editor

Output

Action Output

#	Time	Action	Message
✓ 183	18:26:54	SELECT branch_name, COUNT(*) AS Customer_Count FROM bank_transactions ...	3 row(s) returned
✓ 184	18:26:54	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactio...	1 row(s) returned
✓ 185	18:26:54	SELECT branch_name, AVG(amount) AS Avg_Amount FROM bank_transactions ...	3 row(s) returned
✗ 186	18:26:54	Step 21: WHERE + GROUP BY + HAVING SELECT branch_name, SUM(amount) ...	Error Code: 1064. You have an error in your SQL syntax; check the man...

Step 13: GROUP BY + COUNT()

```

90  -- Step 18: GROUP BY + COUNT()
91  •  SELECT branch_name,
92         COUNT(*) AS Customer_Count
93  FROM bank_transactions
94  GROUP BY branch_name;

```

Result Grid		
	branch_name	Customer_Count
▶	Hyderabad	4
	Vijayawada	3
	Vizag	3

Result 17 x				
Output				
Action Output				
#	Time	Action	Message	
✓ 19	21:38:41	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT...	1 row(s) returned	
✓ 20	22:08:34	SELECT branch_name, MAX(amount) AS Highest_Amount FROM bank_transactions GROUP BY branch...	3 row(s) returned	
✗ 21	22:09:54	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions LIMIT 0, 1000	Error Code: 1140.	
✓ 22	22:10:54	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions GROUP BY branch_...	3 row(s) returned	
✓ 23	22:13:49	SELECT * FROM bank_transactions WHERE amount > 8000 ORDER BY amount DESC LIMIT 0, 1000	4 row(s) returned	
✓ 24	22:14:45	SELECT branch_name, COUNT(*) AS Customer_Count FROM bank_transactions GROUP BY branch_na...	3 row(s) returned	

Step 14: HAVING + COUNT()

```

95  -- Step 19: HAVING + COUNT()
96  • SELECT branch_name,
97        COUNT(*) AS Total_Transactions
98  FROM bank_transactions
99  GROUP BY branch_name
100 HAVING COUNT(*) > 3;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
branch_name	Total_Transactions		
Hyderabad	4		

Result 18 x

Output

Action Output

#	Time	Action	Message
✓ 20	22:08:34	SELECT branch_name, MAX(amount) AS Highest_Amount FROM bank_transactions GROUP BY branch...	3 row(s) returned
✗ 21	22:09:54	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions LIMIT 0, 1000	Error Code: 1140. In
✓ 22	22:10:54	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions GROUP BY branch_...	3 row(s) returned
✓ 23	22:13:49	SELECT * FROM bank_transactions WHERE amount > 8000 ORDER BY amount DESC LIMIT 0, 1000	4 row(s) returned
✓ 24	22:14:45	SELECT branch_name, COUNT(*) AS Customer_Count FROM bank_transactions GROUP BY branch_na...	3 row(s) returned
✓ 25	22:15:40	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_...	1 row(s) returned

Step 15: GROUP BY + AVG()

```

101  -- Step 20: GROUP BY + AVG()
102  • SELECT branch_name,
103        AVG(amount) AS Avg_Amount
104  FROM bank_transactions
105  GROUP BY branch_name;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
branch_name	Avg_Amount		
Hyderabad	4875.000000		
Vijayawada	8000.000000		
Vizag	9000.000000		

Result 19 x

Output

Action Output

#	Time	Action	Message
✗ 21	22:09:54	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions LIMIT 0, 1000	Error Code: 1140. In
✓ 22	22:10:54	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions GROUP BY branch_...	3 row(s) returned
✓ 23	22:13:49	SELECT * FROM bank_transactions WHERE amount > 8000 ORDER BY amount DESC LIMIT 0, 1000	4 row(s) returned
✓ 24	22:14:45	SELECT branch_name, COUNT(*) AS Customer_Count FROM bank_transactions GROUP BY branch_na...	3 row(s) returned
✓ 25	22:15:40	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_...	1 row(s) returned
✓ 26	22:16:45	SELECT branch_name, AVG(amount) AS Avg_Amount FROM bank_transactions GROUP BY branch_na...	3 row(s) returned

Step 16: WHERE + GROUP BY + HAVING

```
106  -- Step 21: WHERE + GROUP BY + HAVING
107  • SELECT branch_name,
108         SUM(amount) AS Total_Deposit
109  FROM bank_transactions
110  WHERE transaction_type = 'Deposit'
111  GROUP BY branch_name
112  HAVING SUM(amount) > 15000;
```

Result Grid

	branch_name	Total_Deposit
▶	Vijayawada	23000.00
	Vizag	23000.00

Result 20 x

Output

#	Time	Action	Message
✓ 22	22:10:54	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions GROUP BY branch_...	3 row(s) returned
✓ 23	22:13:49	SELECT * FROM bank_transactions WHERE amount > 8000 ORDER BY amount DESC LIMIT 0, 1000	4 row(s) returned
✓ 24	22:14:45	SELECT branch_name, COUNT(*) AS Customer_Count FROM bank_transactions GROUP BY branch_na...	3 row(s) returned
✓ 25	22:15:40	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_...	1 row(s) returned
✓ 26	22:16:45	SELECT branch_name, AVG(amount) AS Avg_Amount FROM bank_transactions GROUP BY branch_na...	3 row(s) returned
✓ 27	22:17:40	SELECT branch_name, SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_t...	2 row(s) returned

Step 17: GROUP BY + HAVING + ORDER BY


```

113 -- Step 22: GROUP BY + HAVING + ORDER BY
114 • SELECT branch_name,
115        AVG(amount) AS Avg_Amount
116 FROM bank_transactions
117 GROUP BY branch_name
118 HAVING AVG(amount) > 7000
119 ORDER BY Avg_Amount DESC;

```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	branch_name	Avg_Amount			
▶	Vizag	9000.000000			
	Vijayawada	8000.000000			

Result 21 x

Output				
Action Output				
#	Time	Action	Message	
✓ 23	22:13:49	SELECT * FROM bank_transactions WHERE amount > 8000 ORDER BY amount DESC LIMIT 0, 1000	4 row(s) returned	
✓ 24	22:14:45	SELECT branch_name, COUNT(*) AS Customer_Count FROM bank_transactions GROUP BY branch_na...	3 row(s) returned	
✓ 25	22:15:40	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_...	1 row(s) returned	
✓ 26	22:16:45	SELECT branch_name, AVG(amount) AS Avg_Amount FROM bank_transactions GROUP BY branch_na...	3 row(s) returned	
✓ 27	22:17:40	SELECT branch_name, SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_t...	2 row(s) returned	
✓ 28	22:18:22	SELECT branch_name, AVG(amount) AS Avg_Amount FROM bank_transactions GROUP BY branch_na...	2 row(s) returned	